

Taming infinity

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July 21, 2025

People do mathematics for many different reasons, usually either to achieve some practical purpose or to satisfy curiosity. This has been the case as far back as we have records - ancient Babylonian tablets used for business accounting, which seem to contain the solution of some quadratic equations, and Euclid's geometry, which goes far beyond any obvious applications.

I'm starting with this banal observation because I'm not really sure how to explain why I've spent half a century studying groups. Actually the choice of topic was pretty much an accident, this just happened to be what my PhD supervisor did. But I'll try to give a feeling of why it might be interesting.

Groups are an algebraic concept, the formal descriptors of *symmetry*. As such they play a key role in many parts of physics, where symmetry is a powerful unifying principle (it underlies conservation laws). To mention two examples (just to impress you with some names - these are not topics I know much about)

- the geometry of Einstein's special relativity is determined by the group of Lorentz transformations, while the space of Newtonian mechanics is the familiar Euclidean space, with the affine group as symmetry group.
- the 'standard model' of particle physics associates different particles to complex representations of the unitary groups $U(1)$, $SU(2)$ and $SU(3)$.

So groups are certainly useful - I believe that satnavs wouldn't work without special relativity, and nuclear physics is based on the theory of fundamental particles (whether this knowledge is more dangerous than useful is a good question...).

But once an idea has crystallized, what mostly drives mathematicians is *curiosity*: in my case anyway. What do these things look like? How different can they be? How do they relate to each other, and to other bits of mathematics? Questions seem to present themselves, and if you can answer one of them it is a particular kind of satisfaction. Years ago a philosopher friend said to me 'I really envy mathematicians: you get *results*'. Once you've proved something it's there for good, no-one can take it away from you.

In fact the concept of a group grew out of pure curiosity. It was probably first articulated by the teenager Evariste Galois in the 1820s. He wanted to understand the solution of algebraic equations, and his observation of genius was that this process is in essence the decomposition of a certain group, now

called the Galois group. This is such a key idea I want to show you what it is. The equation

$$X^4 + X^3 + X^2 + X + 1 = 0$$

has 4 solutions, the primitive 5th roots of unity

$$\zeta_i = e^{2\pi i/5} \quad (i = 1, 2, 3, 4).$$

These are not independent: $\zeta_2 = \zeta_1^2$, $\zeta_3 = \zeta_1^3$ etc. Now the Galois group is the set G of all those permutations of $\{\zeta_1, \zeta_2, \zeta_3, \zeta_4\}$ that preserve the relations which hold between these roots. So

$$(1234) \in G$$

$$(12) \notin G$$

for example: we can't swap ζ_1 and ζ_2 because $\zeta_2 = \zeta_1^2$ but $\zeta_1 \neq \zeta_2^2$. In fact G is just the cyclic group of order 4 generated by (1234) . Galois's insight was that this is the special feature that makes it easy to solve the equation; in fact he proved that an equation can be solved 'by radicals' - rational expressions involving n th roots - if and only if the corresponding group is 'easy', what is now called *soluble*; this means that inside the group there are no 'hard bits'. These hard bits are called *finite simple groups*, and I will talk about these later.

If anyone here doesn't already know Galois theory, let me say: *do learn this beautiful theory!* It is a perfect example of what, to me, makes mathematics so satisfying: by giving us the power to do the apparently impossible it turns us - momentarily - into gods. There are infinitely many algebraic equations. Give me any one of them; I can tell you its solutions if any exist, if none exist I will tell you that. How? By a finite calculation of the Galois group. (*In principle*....it might take a LOT of computing. But the point is, it's a finite computation.)

A famous open problem is this: which finite groups actually occur as a Galois group? (Over the *rational*s, for experts - every finite group is a Galois group over some field.) If you can prove that every group does, or find a single one that definitely doesn't, you will probably win a Fields Medal - you certainly ought to!

In general, a group is a nonempty set with a binary operation that satisfies some simple conditions. A better way to view it is this. Take an object S - which can be abstract, like the set $\{1, 2, 3, 4\}$. The set of all permutations of S - bijective mappings $S \rightarrow S$ - is called $\text{Sym}(S)$, the symmetric group on S . The binary operation is concatenation - the point is that these can be composed, each permutation has an inverse, and (a bit technical) the associative law holds: $(ab)c = a(bc)$. Now: a group is any subset of some symmetric group which is closed under the binary operation and inversion.

The *order* of a group is the number of elements (it could be an infinite cardinal).

We usually consider two groups to be the same if they are *isomorphic*: there is a bijection between them that preserves the binary operation. In this sense, there is just one group of order 1, denoted 1, and one of order p for each prime p , denoted C_p (for 'cyclic').

Thus we think of the Galois group G above as a subgroup of $\text{Sym}(4) = \text{Sym}(\{1, 2, 3, 4\})$.

If S is an infinite set then $\text{Sym}(S)$ is infinite. So lots of groups are infinite, such as the Lorentz group and the unitary groups mentioned above. Actually these are what used to be called ‘continuous groups’, nowadays *Lie Groups*. These groups are not just algebraic structures: they also have a *topology* - each point has a neighbourhood that looks like a little patch of Euclidean space.

Here is $U(1)$ for example **Draw circle, mention group of rotations and show a co-ordinate patch**

I’ll come back to topological groups later.

A different style of infinite group is *countable*. The basic model here is the additive group of integers $\mathbb{Z}$. This is a discrete subgroup of $\mathbb{R}$; an important part of twentieth-century mathematics was devoted to understanding to what extent the discrete subgroups of a Lie group control the structure of the Lie group - another glimpse of what I mean by ‘taming the infinite’. Here the ‘small’ (countable) subgroups give strong information about the (uncountable) Lie group. For cognoscenti - I’m talking here about arithmetic subgroups of semisimple Lie groups, the basic example is

$$\text{SL}_n(\mathbb{Z}) < \text{SL}_n(\mathbb{R}).$$

The background to this is the discovery and eventual classification of the *semisimple Lie groups*, invented by Sophus Lie and ultimately classified by Killing and Cartan. What they discovered was that each of these groups was completely specified by a ‘root system’, I won’t define this, except to say that it’s a little star-like configuration of finitely many vectors in a finite-dimensional real vector space; and the amazing fact is that there are only a small (finite) number of such configurations in each dimension. For example, the root system corresponding to $\text{SL}_2(\mathbb{R})$ is one-dimensional

$$A_1: \quad \longleftarrow \bullet \longrightarrow$$

it is the only one, but in two dimensions there are 3 (indecomposable) ones which I will draw

Now an extraordinary thing happened. Over the course of the last century, it emerged that there is a whole universe of *finite* groups that mirrors this family of Lie groups. These finite simple groups are what I called the ‘hard bits’ in Galois theory. Galois himself studied some of them in detail, such as $\text{SL}_2(\mathbb{F}_p)$, a group of 2×2 matrices over the finite field of size p , an odd prime (this is not exactly simple but close enough). Anyway, for each root system R and each field F Claude Chevalley showed how to make a group $R(F)$, so that when F is $\mathbb{R}$ or $\mathbb{C}$ we get back the Lie group, while if F is a finite field we get a finite

group; and these groups are all (essentially) simple. For example

$$\begin{aligned} A_1(\mathbb{R}) &= \mathrm{SL}_2(\mathbb{R}) \\ A_1(\mathbb{F}_p) &= \mathrm{SL}_2(\mathbb{F}_p). \end{aligned}$$

I haven't actually said what 'simple group' means. Conceptually, a group is simple if it can't be decomposed into smaller bits. This means it has *no proper homomorphic image* (like a prime number: the only normal subgroups are $\{1\}$ and the whole group). The dream of group theorists is to understand the structure of groups; the way to get there is to decompose the group as far as possible into smaller - hopefully easier - pieces. But this process grinds to a halt when we get to a piece that can't be decomposed any further. That's a simple group.

So the big - and one would have said certainly impossible - dream was to 'know' all the simple groups (among finite groups, that is). As Galois already knew, the basic examples are matrix groups like $\mathrm{SL}_n(\mathbb{F}_p)$ and also the alternating groups $\mathrm{Alt}(n)$ (which is nearly the whole of $\mathrm{Sym}(n)$). Well to cut a long story very short, a towering achievement of 20th century mathematics

(one of them - the other was Grothendieck's reinvention of algebraic geometry. Incidentally, Alexander Grothendieck recognised in Galois his only equal among historical mathematicians!)

was the

Classification of finite simple groups.

This enormous theorem (15,000 pages of proof) says that every finite simple group is in the following list:

- alternating groups
- Chevalley groups (and 'twisted' versions')
- 26 sporadic groups.

This is 'taming infinity' in a spectacular way. Like saying that there are 5 Platonic solids in 3 dimensions, this says that in each 'dimension' (in some technical sense), there are just finitely many basic finite symmetrical structures, and it gives a formula to construct each one.

But for me, to tame the infinite means to reduce it to the *finite*. What could this mean? This kind of mathematics began in number theory, perhaps with Fermat, who showed that for certain equations, the existence of an integer solution depends on whether the coefficients of the equation satisfy some congruences:

$$X^2 + Y^2 = p \tag{1}$$

for an odd prime p is solvable if and only if $p \equiv 1 \pmod{4}$. In a sense this is still a finite problem since any solution must satisfy $|X| < p$, $|Y| < p$ (though

it's certainly easier to find the remainder when you divide p by 4 than to check out all squares less than p); but there are similar results for

$$X^2 - dY^2 = p, \quad (2)$$

at least for 'nice' values of d , for example $d = 2$ (to do with whether d is congruent to a square modulo p); the point is that in this case there is no a priori bound on the size of possible solutions, so it is really an 'infinite' problem: and Fermat reduces it to a finite question.

If the answer to this question is 'yes', then the equation actually has infinitely many solutions: but we can describe them all at once by noting that they form exactly one orbit of a certain group (in this case a very 'simple' group) - I'll come back to this later.

Gauss and Legendre extended these ideas to quadratic forms in 3 variables. In all cases, the idea is that an equation really can be solved in integers if it can be solved 'modulo n ' for all n (and in practice one only has to check a finite number of ns).

In the 20th century, Hasse and Minkowski proved the most general version, which essentially says the same thing for all quadratic forms. The idea is this:

- certain properties of $\mathbb{Z}$ (the ring) are determined by properties of the finite rings $\mathbb{Z}/n\mathbb{Z}$.

So to answer some question about the infinite thing $\mathbb{Z}$, we need to ask infinitely many questions about finite things (the rings $\mathbb{Z}/n\mathbb{Z}$). This doesn't look like much progress. But in mathematics we like to construct an object that can capture a lot of information in one box, as it were. One way to do this is form an *inverse limit*.

Fix a prime p and consider the family

$$\{\mathbb{Z}/p^n\mathbb{Z} \mid n = 1, 2, \dots\}.$$

Each of these little groups maps onto the preceding one in a natural way:

$$\pi_n : a \bmod p^n \longmapsto a \bmod p^{n-1}.$$

Now form the Cartesian product

$$P := \prod_{n=1}^{\infty} \mathbb{Z}/p^n\mathbb{Z}$$

and let

$$G = \{(b_1, b_2, \dots) \in P \mid \pi_n(b_n) = b_{n-1} \text{ for each } n > 1\}.$$

This G is a subgroup of P , and G contains a copy of $\mathbb{Z}$: identify $a \in \mathbb{Z}$ with the 'vector' $(a \bmod p^n)_{n \in \mathbb{N}}$. The important thing is that the finite images of G are exactly the groups $\mathbb{Z}/p^n\mathbb{Z}$, and these are just those finite images of $\mathbb{Z}$ whose order is a power of p .

This G is called the ring of *p-adic integers*, denoted

$$G = \mathbb{Z}_p.$$

In what way does this big - uncountable! - object tame the infinitude of $\mathbb{Z}$? To cut a long story short, G has a *topology*, and as such it is a compact topological group. (For experts: G is a closed subspace of P which has the product topology, and P is compact by Tychonoff's Theorem.)

Now 'compact' is kind of like 'finite' in the world of topology

draw a circle and a line

An the upshot is that we can express infinitely many facts - about the finite things $\mathbb{Z}/p^n\mathbb{Z}$ - as a single fact about the compact thing $\mathbb{Z}_p$.

Roughly speaking, the Hasse-Minkowski Theorem says that an integral quadratic form represents zero (nontrivially) over $\mathbb{Z}$ if and only if it does so over $\mathbb{Z}_p$ for every prime p .

This construction works for any sequence of finite groups with 'backward mappings'. We define

- a *profinite group* is an inverse limit of a sequence (more precisely, an 'inverse system') of finite groups.

So a profinite group is (a special kind of) compact topological group. It is very different from the familiar Lie groups mentioned before: instead of being 'continuous' it is 'totally discontinuous'. I won't define this, but it does mean that one's topological intuitions, based on visualising Euclidean space, can be misleading.

It turned out that profinite groups arise naturally, in a quite different guise, in number theory: unlike the p -adics these are not (usually) commutative. To see these, we need to recall the more modern description of a Galois group as the *automorphism group of a field extension*. If the extension is finite this is indeed the Galois group of an equation. In general, this group is exactly the inverse limit of the Galois groups of all the finite subextensions (for experts: this is simply saying no more nor less than that a field automorphism is determined by its action on each element!). This observation was made by Wolfgang Krull in the early years of the last century; he recast Galois theory for arbitrary (possibly infinite) field extensions as a correspondence between intermediate extension fields and closed subgroups of a profinite group.

Now the question I mentioned before can be phrased in the following - equivalent - way:

- What is the absolute Galois group of $\mathbb{Q}$?

That is, the group of all automorphisms of the field of all algebraic numbers; it's a profinite group.

Back to group theory, what is the relation between $\mathbb{Z}$ and $\mathbb{Z}_p$? In general, for any group G we can consider the family $\mathcal{F}$ of all the finite quotients of G , and form its inverse limit, denoted $\widehat{G}$. This is the *profinite completion* of G . It is a single, compact, group, that carries exactly all the information about G that can be seen in its finite images. In fact $\mathbb{Z}_p$ is not exactly the profinite

completion of $\mathbb{Z}$ because it depends only on the p -power sized images of $\mathbb{Z}$. The Chinese Remainder Theorem says exactly that

$$\widehat{\mathbb{Z}} = \prod_{p \text{ prime}} \mathbb{Z}_p.$$

Now we can ask: *do the finite images of a group G determine the group?* This is the same as saying that G is determined by $\widehat{G}$. Usually the answer is ‘no’, for many reasons, some obvious some less obvious. For example, the additive group of $\mathbb{Q}$ has only the trivial finite image...So it’s sensible to consider only *residually finite* groups: G is residually finite if distinct elements of G remain distinct in some finite image.

Now this question is rather like the number theory questions I mentioned before: sometimes the solvability an equation in $\mathbb{Z}$ is determined by its solvability ‘modulo n ’ for many integers n . A related concept is *equivalence*, for quadratic forms, say. I won’t define this, but equivalent forms have basically the same properties. Two forms that are ‘equivalent modulo n ’ for every n are said to belong to the same *genus*. A classic result is that each genus consist of *finitely many* equivalence classes.

In the world of groups we imitate this idea: say that groups G and H are in the same *profinite genus* if they have the same finite images - i.e. if $\widehat{G} \cong \widehat{H}$. Usually we only allow groups that are *residually finite* and *finitely generated*. Could it be that a genus of groups consists of finitely many isomorphism classes? Usually no, but it does happen in interesting cases. It is easy to see (if you know a little group theory) that any finitely generated abelian group is determined - *among finitely generated residually finite groups* - by its profinite completion. Such groups are said to be *profinely rigid*. Slightly more complicated groups may not be profinitely rigid, but still have finite genus; these are the polycyclic groups, which are built out of the additive group $\mathbb{Z}$ in finitely many steps, but may be non-commutative. Here is the simplest one:

$$H = \begin{bmatrix} 1 & \mathbb{Z} & \mathbb{Z} \\ & 1 & \mathbb{Z} \\ & & 1 \end{bmatrix}. \quad (3)$$

Theorem 1 (Grunewald, Pickel, Segal, 1980) *The profinite genus of a polycyclic group is finite.*

(This depends on the nontrivial fact that ‘being polycyclic’ is a profinite property, i.e. that the profinite genus of a polycyclic group consists of polycyclic groups (Sabbagh, Wilson, 1991)). More recently (2011) Menny Aka showed that for most *arithmetic groups* the ‘relative profinite genus’ is finite - that is, such a group is determined *among arithmetic groups* up to finitely many possibilities by its profinite completion. Still more recently, Martin Bridson and his collaborators have found many examples of groups that are truly profinitely rigid (among all finitely generated residually finite groups). One of them is

$$\mathrm{PSL}_2(\mathbb{Z}[\omega])$$

where $\omega = e^{2\pi i/3}$. A famous open problem is ‘Remeslennikov’s question’

- *Are finitely generated free groups profinitely rigid?*

These groups are precisely the finitely generated subgroups of

$$F_2 = \left\langle \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix} \right\rangle.$$

In one sense they are the simplest possible groups, in that their generators satisfy no non-trivial relations (as an exercise you could prove this for these two matrices!).

Another aspect of taming infinity is the field of *decision problems*. To say that a certain question is decidable means there is an *algorithm* that tells you the answer, in *finitely many steps*: that is what algorithm means! Simple topological questions about loops in a space, for example, can be formulated as questions about the *fundamental group* of the space. If we had longer I would like to explain what this is, and how one should consider it as a group of transformations. In 1910 Max Dehn was motivated by topology to formulate three basic decision problems about a finitely presented group $G = \langle x_1, \dots, x_d \rangle$

- the *word problem*; given two words u, v , is $u(x_1, \dots, x_d) = v(x_1, \dots, x_d)$?
- the *conjugacy problem*; given two words u, v , does there exist $g \in G$ such that $g^{-1} \cdot u(x_1, \dots, x_d) \cdot g = v(x_1, \dots, x_d)$?
- the *isomorphism problem*; given another finitely presented group H , is G isomorphic to H ?

Famous work by Novikov, Boone, Adian and Rabin in the 1950s showed that these are all *undecidable* in general. So it was a challenge to find particular classes of groups where they are decidable.

One quite general approach is to study a group by its finite images. Let me actually show you a proof!

Theorem 2 *In a finitely presented residually finite group the word problem is solvable.*

Proof. (Sketch) We run two procedures simultaneously.

- Procedure A: list consequences of the relations in the finite presentation of G
- Procedure B: list homomorphisms $f : G \rightarrow Q$ for finite groups Q .

These can be done effectively. I claim that after finitely steps one of the following things happens:

- A*: $u(x_1, \dots, x_d) = v(x_1, \dots, x_d)$ appears as a consequence of the relations
- B*: $f(u(x_1, \dots, x_d)) \neq f(v(x_1, \dots, x_d))$ for some homomorphisms f from G to some finite group.

If A* happens we conclude that $u(x_1, \dots, x_d) = v(x_1, \dots, x_d)$. If B* happens we conclude that $u(x_1, \dots, x_d) \neq v(x_1, \dots, x_d)$.

So either way we have answered the question in finitely many steps. ■

Could a similar idea work for the other problems? Well, in principle yes, but the group would need to be rather special. For example, by the same argument (essentially) we have

Theorem 3 *In the class of finitely presented profinitely rigid groups the isomorphism problem is solvable.*

In the 1970s, Fritz Grunewald and I had to work much harder to establish

Theorem 4 (1980) *The conjugacy problem is solvable in every arithmetic group.*

Theorem 5 (1980 - 1990) *The isomorphism problem is solvable in the class of polycyclic groups.*

We recall that the profinite genus of a polycyclic group is *finite*, but this is not enough! (If one could determine the exact size of the genus, one could then manage with a hand-wavy procedure as above - but this problem seems to be just as difficult.)

What lies behind these results is a far-reaching generalization of the theory of quadratic forms, based on the theory of *algebraic groups*. I won't define these, they are groups of matrices that arise in a fairly natural way. They include things like the Chevalley groups mentioned before.

Now we specialize to what are called *arithmetic groups*. These are the 'integer points' of algebraic groups, for example the 'Heisenberg group' (3) above, and the group

$$\Gamma = \mathcal{G}(\mathbb{Z}) = \mathrm{GL}_d(\mathbb{Z})$$

which is an arithmetic subgroup of the algebraic group

$$\mathcal{G} = \mathrm{GL}_d.$$

Now a matrix group acts on a vector space in a natural way, and we can think about *orbits* - vectors a and b are in the same orbit of the group Γ if there exists $\gamma \in \Gamma$ such that $a\gamma = b$. If everything has entries in $\mathbb{Z}$ we can also consider this equation modulo n . I'll say that a and b are in the same *local orbit* if for each $n \neq 0$ there exists $\gamma = \gamma(n) \in \Gamma$ such that $a\gamma \equiv b \pmod{n}$.

Theorem 6 (Borel, Serre, 1960s) *Let Γ be an arithmetic group acting rationally on $V = \mathbb{Z}^d$. Then every local orbit of Γ is the union of finitely many orbits.*

Theorem 7 (Grunewald, Segal, 1980) *Let Γ be an arithmetic group acting rationally on $V = \mathbb{Z}^d$. There is an algorithm that decides for any two points of V whether they belong to the same orbit of Γ .*

The first of these encompasses many ‘finiteness theorems’ in number theory; the second encompasses many decision problems in algebra.

We proved a companion result

Theorem 8 (Grunewald, Segal, 1980) *Let Γ be an arithmetic group. There is an algorithm that finds a finite generating set for Γ .*

Here, Γ is given by an *arithmetic* definition, such as

$$\mathrm{SL}_d(\mathbb{Z}) = \{A \in \mathrm{M}_n(\mathbb{Z}) \mid \det A = 1\},$$

and the algorithm produces finitely many actual matrices that generate this as a group. This is key to another kind of taming infinity: some equations, like

$$X^2 - 3Y^2 = 13, \tag{4}$$

have infinitely many solutions, but we can list all the solutions in a finite way by using *the symmetry group of the equation*, in this case the group of units of the ring $\mathbb{Z}[\sqrt{3}]$, the set

$$\begin{aligned} U &= \{x + y\sqrt{3} \mid x^2 - 3y^2 = 1\} \\ &= \langle -1, 2 + \sqrt{3} \rangle \end{aligned}$$

which says that the group U is generated by -1 and just one more element, the ‘fundamental unit’, $2 + \sqrt{3}$. We can think of this acting on the solutions (a, b) as a 2×2 matrix:

$$(a + b\sqrt{3})(2 + \sqrt{3}) = a' + b'\sqrt{3}$$

where $(a', b') = (a, b)M$ and

$$M = \begin{pmatrix} 2 & 1 \\ 3 & 2 \end{pmatrix}.$$

The fact is that the *complete set of (integer) solutions* of (4) is

$$\pm(4, 1)M^n, n \in \mathbb{Z}.$$

This sort of thing happens in many situations. The **Dirichlet Units Theorem** (1846) provides the same kind of answer for all so-called ‘norm form’ equations, in several variables. It describes (more or less explicitly) a finitely generated group, such that the solutions make up finitely many orbits of this group. In the 20th century, L. J. Mordell proved that the set of all *rational* solutions of many cubic equations in two variables (‘elliptic curves’) forms a finitely generated (abelian) group; this was the beginning of modern ‘arithmetic geometry’

Using Theorem 8 (taking for Γ the orthogonal group over $\mathbb{Z}$ of some quadratic form), one obtains

Theorem 9 (Grunewald, Segal, 1981; C-L. Siegel, 1972) *There is an algorithm that decides for an arbitrary quadratic equation (over $\mathbb{Z}$) if it has infinitely many, finitely many, or no solutions.*

The algorithm in principle describes all the solutions in terms of finitely many orbits of an explicitly finitely generated matrix group.

While maybe not too exciting, this is interesting because of **Hilbert’s 10th problem**, famously proved to be unsolvable in general by Matiyasevic et al in 1970. This asks for an algorithm to decide whether *an arbitrary polynomial equation over $\mathbb{Z}$ has any solutions*. No such algorithm exists for equations of degree 4. The case of degree 3 is unknown!

If time remains

A different aspect of taming the infinite arises if we restrict the language we are allowed to use. First-order logic is good for logicians but bad for most mathematicians, because we are not allowed to talk about subsets of things. If you are a group theorist you might be interested in questions like this: *can you specify a group by its first-order properties?*

A group G is *axiomatizable* if every group that satisfies the same first-order sentences as G is isomorphic to G .

It’s obvious that a finite group is axiomatizable: just write down the multiplication table. In fact such a group is *finitely axiomatizable*: one sentence suffices (the conjunction of finitely many sentences).

For infinite groups, it only makes sense to ask if a group is axiomatizable subject to some set-theoretical constraints, since any first-order theory that has an infinite model has models of arbitrary infinite cardinalities (Löwenheim-Skolem Theorem). For example, one looks for groups that are axiomatizable, or finitely axiomatizable, in the class of all finitely generated groups. A famous and very difficult result is

Theorem 10 (Sela, Kharlampovich-Miasnikov, 2006) *All finitely generated non-abelian free groups satisfy the same first-order sentences.*

So the free group F_2 is far from axiomatizable. This made the next result rather surprising.

One would expect first-order logic to be even weaker for talking about topological groups, since topological concepts just can’t be expressed in this language. However, profinite groups seem to behave rather like finite groups in this context:

Theorem 11 (Jarden, Lubotzky, 2008) *Every finitely generated profinite group is axiomatizable (among profinite groups).*

In particular, in contrast to the case of ‘abstract’ free groups we see that each of the ‘free profinite groups’ $\widehat{F}_n$ is uniquely determined by its first-order theory.

What makes this possible is

Theorem 12 (Nikolov, Segal, 2007). *Let G be a finitely generated profinite group. Then*

1. *For each n , every product of n th powers in G is equal to a product of $f(n)$ n th powers.*
2. *Every subgroup of finite index in G is open and definable.*
3. *The topology on G is completely determined by the group structure.*
4. *Every group homomorphism from G to any profinite group is continuous.*

The key result is **1.**, the rest are easy consequences. The key point for axiomatizability is **2.**: if the subgroups of finite index are definable then the finite quotient groups can be specified, and these determine the profinite group (which is equal to their inverse limit).

Theorem 12 takes about 100 pages to prove, and it depends strongly on CFSG. This is pretty remarkable: knowing the detailed structure of all the finite simple groups is key to some very general results in what looks like a completely different area.

Now if I want to claim that I have tamed infinity I need to find out: which finitely generated (abstract or profinite) groups are *finitely* axiomatizable? There are several negative results of a general nature, and a few positive results. A brief sample:

Theorem 13 (Oger, Sabbagh, 2006) *Groups like the Heisenberg group H (3) are finitely axiomatizable (among finitely generated groups), the group $H \times \mathbb{Z}$ is not.*

Theorem 14 (Nies, Segal, Tent, 2021) *Groups like $H(\mathbb{Z}_p)$ are finitely axiomatizable (among profinite groups), the group $\hat{H}$ is not.*

Theorem 15 (Segal, Tent, 2022) *Let $\mathcal{G}$ be an adjoint simple Chevalley group scheme of rank at least 2. Then $\mathcal{G}(\mathbb{Z})$ and $\mathcal{G}(\mathbb{F}_p[t])$ are finitely axiomatizable among finitely generated groups; $\mathcal{G}(\mathbb{Z}_p)$ and $\mathcal{G}(\mathbb{F}_p[[t]])$ are finitely axiomatizable among profinite groups, while $\mathcal{G}(\hat{\mathbb{Z}})$ is not.*

These groups include the ‘classical groups’ like $\mathrm{PSL}_n(-)$, $n \geq 3$.